



Problem Solving With C
UE24CS151B

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1. **(1 Mark):** What is an identifier in C programming?
2. **(1 Mark):** What keyword is used to declare an integer variable in C?
3. **(2 Marks):** Explain the difference between declaring and initializing a variable in C.
4. **(2 Marks):** Provide two examples of invalid identifiers in C and explain why they are invalid.
5. **(3 Marks):** What are keywords in C programming, and why can't they be used as variable names? Give three examples of keywords.
6. **(3 Marks):** Describe the purpose of the `char` data type in C and provide an example of how it is used.
7. **(4 Marks):** Explain the difference between the `float` and `double` data types in C. When would you choose to use `double` over `float`?
8. **(5 Marks):** Explain the concept of scope of a variable in C. Describe at least two different types of scope.
9. **(5 Marks):** Discuss the significance of data types in C. How do data types affect memory allocation and the operations that can be performed on variables?
10. **(10 Marks):** Imagine you need to store a student's name, age, and GPA in a C program.